

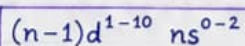
d- AND f-BLOCK ELEMENTS

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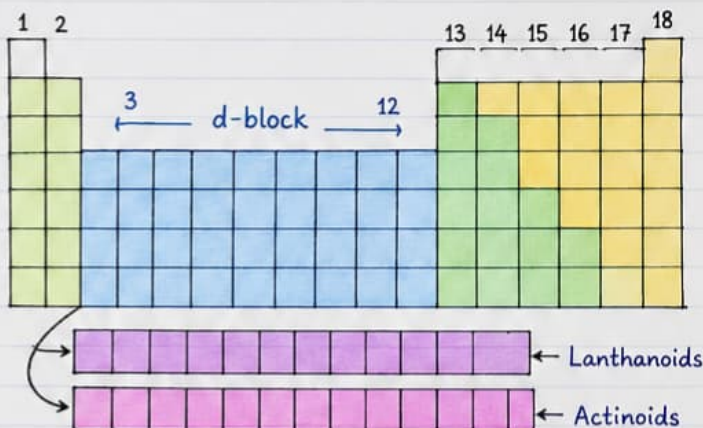
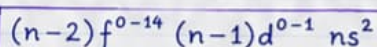
INTRODUCTION

- The elements in which the differentiating electron enters $(n-1)d$ or $(n-2)f$ subshell are called **d-block** and **f-block** elements.

- d-block elements → Groups 3 to 12



- f-block elements → Inner transition elements (Lanthanoids & Actinoids)



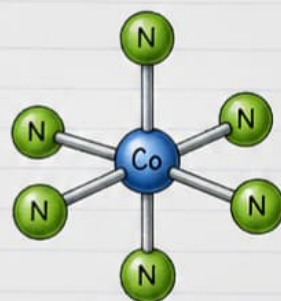
GENERAL PROPERTIES OF d-BLOCK ELEMENTS

- Variable oxidation states → Due to small energy difference between $(n-1)d$ and ns orbitals.
- Formation of coloured compounds → Because of d-d transitions.
- Paramagnetism → Due to unpaired d-electrons.
- Catalytic activity → Many d-block elements and their compounds act as catalysts.
- Formation of complex compounds → Due to small size and high charge density.

d-electrons take part in bonding, so their compounds show interesting properties.

ELECTRONIC CONFIGURATION & EXAMPLES

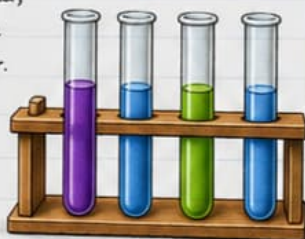
Element	Atomic No.	Electronic Configuration	Block	Differentiating Electron
Scandium (Sc)	21	$[Ar] 3d^1 4s^2$	d-block	$3d^1$
Zinc (Zn)	30	$[Ar] 3d^{10} 4s^2$	d-block	$3d^{10}$
Lanthanum (La)	57	$[Xe] 5d^1 6s^2$	d-block (outer)	$5d^1$
Cerium (Ce)	58	$[Xe] 4f^1 5d^1 6s^2$	f-block	$4f^1$



Octahedral complex
 $[Co(NH_3)_6]^{3+}$

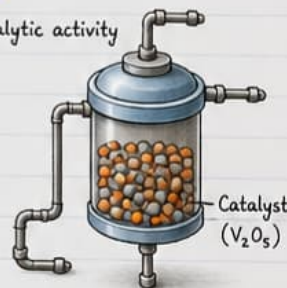
COLOUR IN COMPOUNDS

- When light passes through a compound, some wavelengths are absorbed and the remaining are transmitted → we see colour.
- Example:
 - $[Ti(H_2O)_6]^{3+}$ → Violet
 - $[V(H_2O)_6]^{2+}$ → Blue
 - $[Cr(H_2O)_6]^{3+}$ → Green
 - $[Cu(H_2O)_6]^{2+}$ → Blue



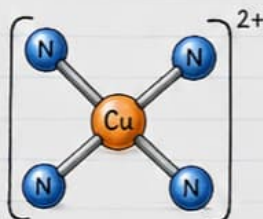
CATALYTIC ACTIVITY

- d-block elements show high catalytic activity due to variable oxidation states.
- Example:
 - Fe_2O_3 in Haber process
 - V_2O_5 in Contact process
 - Ni in hydrogenation of oils



COMPLEX FORMATION

- d-block elements form complex compounds with ligands (Lewis bases).
- Example:
 - $[Cu(NH_3)_4]^{2+}$, $[Fe(CN)_6]^{4-}$
- Usually coordination number = 4 or 6.



MAGNETIC PROPERTIES

- Presence of unpaired d-electrons → Paramagnetic
- No unpaired d-electrons → Diamagnetic
- Example:
 - $[Fe(H_2O)_6]^{2+}$ → Paramagnetic (4 unpaired)
 - $[Zn(H_2O)_6]^{2+}$ → Diamagnetic (0 unpaired)



d- AND f- BLOCK ELEMENTS

CHEMICAL PROPERTIES

1. Variable oxidation states

- d-block elements show variable oxidation states due to small energy difference between $(n-1)d$ and ns orbitals.

Examples

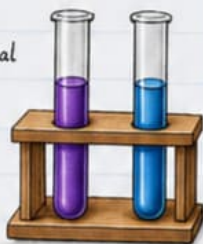
Element	Common Oxidation States	Compounds
Sc	+2, +3	$ScCl_2$, $ScCl_3$
Ti	+2, +3, +4	$TiCl_2$, $TiCl_3$, $TiCl_4$
V	+2, +3, +4, +5	VCl_2 , VCl_3 , $VOCl_3$, VCl_5
Cr	+2, +3, +6	$CrCl_2$, $CrCl_3$, CrO_3
Mn	+2, +4, +6, +7	$MnCl_2$, MnO_2 , MnO_3 , $KMnO_4$

2. Formation of coloured compounds

- Due to d-d transitions (involves partial absorption of visible light).

Example

- $[Ti(H_2O)_6]^{3+} \rightarrow$ Violet
- $[Cu(H_2O)_6]^{2+} \rightarrow$ Blue



3. Complex formation

- d-block elements form complex compounds due to - small size, high charge density, - vacant d-orbitals.

Example: $[Fe(CN)_6]^{4-}$, $[Cu(NH_3)_4]^{2+}$

4. Catalytic activity

- Many d-block elements and their compounds act as catalysts.

Examples

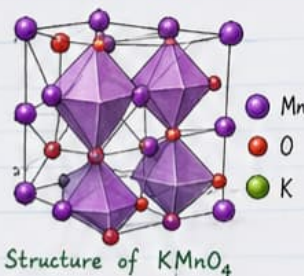
- Fe in Haber process
- V_2O_5 in Contact process
- Ni in hydrogenation of oils
- MnO_2 in decomposition of H_2O_2



SOME IMPORTANT COMPOUNDS

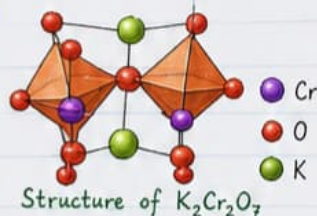
1. $KMnO_4$ (Potassium permanganate)

- Dark purple crystalline solid.
- Strong oxidising agent in neutral or alkaline medium.
- Used in titrations.



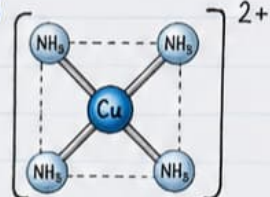
2. $K_2Cr_2O_7$ (Potassium dichromate)

- Orange crystalline solid.
- Strong oxidising agent in acidic medium.
- Used in titrations.



3. $[Cu(NH_3)_4]^{2+}$ (Tetraamminecopper(II) ion)

- Deep blue colour.
- Square planar geometry.
- Example of complex ion.



GENERAL TRENDS IN d-BLOCK ELEMENTS

Property / Trend	Trend Across the Series (3d Series)
Atomic size	Decreases gradually (due to poor shielding by d-electrons)
Ionisation enthalpy	Increases gradually
Oxidation states	Increase from Sc to Mn, then decrease from Mn to Zn
Basic character of oxides	Decreases from Sc_2O_3 (basic) to ZnO (amphoteric)
Magnetic character	Paramagnetic for most, Zn^{2+} is diamagnetic (d^{10})

LANTHANIDES (4f-BLOCK ELEMENTS)

- Inner transition elements with general electronic configuration: $(n-2)f^{0-14} (n-1)d^{0-1} ns^2$
- All trivalent (+3 oxidation state) and silvery-white, soft metals.
- Show gradual decrease in atomic and ionic radii from La^{3+} to Lu^{3+} .
- Lanthanide contraction: Steady decrease in size of lanthanides due to poor shielding by 4f-electrons.

APPLICATIONS OF LANTHANIDES

- $CeO_2 \rightarrow$ Used in catalytic converters.
- $La_2O_3 \rightarrow$ Used in camera lenses and optical glass.
- $Eu^{3+} \rightarrow$ Used in colour TV screens (red phosphor).
- $Gd^{3+} \rightarrow$ Used in MRI (magnetic resonance imaging).
- Pr, Nd $\rightarrow$ Used in high strength magnets.



ACTINIDES (5f-BLOCK ELEMENTS)

- Inner transition elements with general electronic configuration: $(n-2)f^{0-14} (n-1)d^{0-1} ns^2$
- All are metallic and highly radioactive.
- Most have oxidation states +3 to +6.
- Actinide contraction present, but less pronounced than lanthanides.



QUICK RECAP

- d-Block $\rightarrow$ Groups 3-12, variable oxidation states.
- f-Block $\rightarrow$ Inner transition elements (Lanthanides & Actinides).
- Show complex formation, coloured compounds & catalytic activity.
- Many important compounds used in industries & analytical chemistry.



d- AND f- BLOCK ELEMENTS

IMPORTANT POINTS

- d-block elements have $(n-1)d$ orbitals partially filled in their atoms or in one of their ions.
- They form **coloured** compounds.
- They exhibit **variable oxidation states**.
- They form **complex compounds**.

CHEMISTRY OF SOME d-BLOCK ELEMENTS

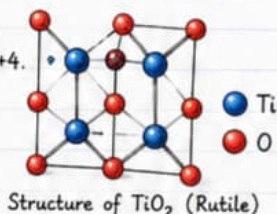
1. Scandium (Sc, Z = 21)

- Shows only +3 oxidation state.
- Forms colourless compounds.
- Example: Sc_2O_3 , ScCl_3



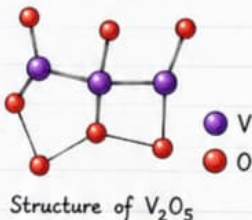
2. Titanium (Ti, Z = 22)

- Common oxidation states: +2, +3, +4.
- TiO_2 is used as white pigment (in paint, paper, plastics).
- Example: TiO_2 , TiCl_4



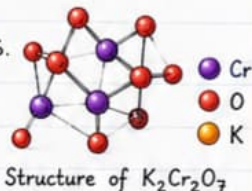
3. Vanadium (V, Z = 23)

- Shows oxidation states from +2 to +5.
- V_2O_5 is a strong oxidising agent.
- Used in manufacture of alloys.
- Example: V_2O_5 , NH_4VO_3



4. Chromium (Cr, Z = 24)

- Important oxidation states: +2, +3, +6.
- Cr_2O_3 is green.
- CrO_3 and $\text{Cr}_2\text{O}_7^{2-}$ are orange.
- Example: $\text{K}_2\text{Cr}_2\text{O}_7$, CrCl_3



TEST YOURSELF

1. Why d-block elements generally show variable oxidation states?
2. Why are most d-block compounds coloured?
3. Give one example each of d-block complex compound.
4. What is the general electronic configuration of f-block elements?



★ EXAM TIP

Remember the trends: ionic size ↓, oxidation state across 3d series & lanthanide contraction! ★

f-BLOCK ELEMENTS (LANTHANIDS & ACTINIDS)

- f-block elements are placed below the main body of the periodic table.
- They are called **inner transition elements**.
- General electronic configuration:

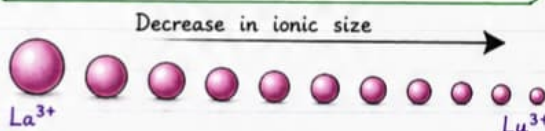
$$(n-2)f^{0-14}(n-1)d^{0-1}ns^2$$

LANTHANIDS

- Elements with filling of 4f subshell.
- Atomic number: 57 (La) to 71 (Lu).
- They show +3 oxidation state (common).
- Exhibit lanthanide contraction.

LANTHANIDE CONTRACTION

The steady decrease in size of lanthanide ions from La^{3+} to Lu^{3+} due to poor shielding by 4f-electrons.

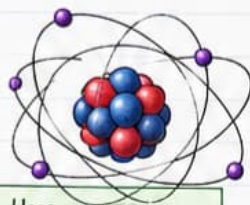


USES OF LANTHANIDS

Element	Uses
La	Used in camera lenses, hybrid car batteries.
Ce	Used in catalytic converters, glass polishing.
Pr	Used in strong magnets, glass colouring.
Nd	Used in laser technology, high strength magnets.
Eu	Used in colour TV screens (red phosphor).

ACTINIDS

- Elements with filling of 5f subshell.
- Atomic number: 89 (Ac) to 103 (Lr).
- Most are radioactive.
- Common oxidation states: +3 to +6.



Element	Oxidation State(s)	Uses
Ac	+3	Starting element of actinoid series.
Th	+4	Used in nuclear reactors.
U	+4, +6	Used as nuclear fuel.
Pu	+3, +4, +5, +6.	Used in nuclear weapons.

SUMMARY



d-block elements show variable oxidation states and form coloured compounds.



They act as good catalysts and form complex compounds.



Trends: ionic size decreases, oxidation state increases across 3d series.



Lanthanides show lanthanide contraction (gradual decrease in ionic size).



Actinoids are radioactive and used in nuclear technology.

CHAPTER 4

d- AND f-BLOCK ELEMENTS

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COLOUR OF d-BLOCK COMPOUNDS

- d-d transitions absorb light in the visible region.
- The complementary colour is observed.

Example:

Ions/Compounds	Absorbed Colour	Observed Colour
$[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$	Violet	Yellow-green
$[\text{V}(\text{H}_2\text{O})_6]^{2+}$	Blue	Green
$[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$	Green	Violet
$[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$	Orange	Blue

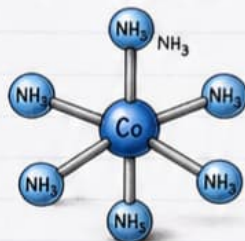
More the number of d-d transitions $\rightarrow$ greater the possibility of colour.



COMPLEXES OF d-BLOCK ELEMENTS

- d-block elements form coordination compounds with ligands (Lewis bases).
- Coordination number is usually 4 or 6.

Example: $[\text{Co}(\text{NH}_3)_6]^{3+}$



- Coordination No. = 6
- Geometry = Octahedral
- Type = Complex ion
- Charge = +3

COMPARISON: d-BLOCK vs f-BLOCK ELEMENTS

Property	d-Block Elements	f-Block Elements
Subshell filled	$(n-1)d$	$(n-2)f$
Position	Groups 3-12	Separate bottom two rows
Nature	Moderately hard and less reactive	Soft and more reactive
Oxidation states	Variable	More variable
Complex formation	Common	Very common
Magnetic nature	Paramagnetic	Usually paramagnetic
Radioactivity	Few (e.g., Tc, Pm)	Most are radioactive

QUICK RECAP

- d-block $\rightarrow$ Groups 3-12, $(n-1)d$ filling.
- f-block $\rightarrow$ Lanthanides (4f) & Actinides (5f).
- Show variable oxidation states & form coloured compounds.
- Exhibit catalytic activity and paramagnetism.
- Form complex compounds with coordination number 4 or 6.
- Lanthanide contraction $\rightarrow$ gradual decrease in size.
- Many actinides are radioactive and used in nuclear technology.

MAGNETIC BEHAVIOUR

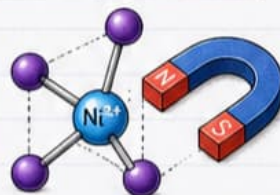
- d-block elements show paramagnetism due to presence of unpaired d-electrons.

Example:

Species	Electronic Configuration	Unpaired e^-	Behaviour
Sc (Z=21)	$[\text{Ar}] 3d^1 4s^2$	1	Paramagnetic
Ti^{2+}	$[\text{Ar}] 3d^2$	2	Paramagnetic
Ni^{2+}	$[\text{Ar}] 3d^8$	2	Paramagnetic
Zn^{2+}	$[\text{Ar}] 3d^{10}$	0	Diamagnetic

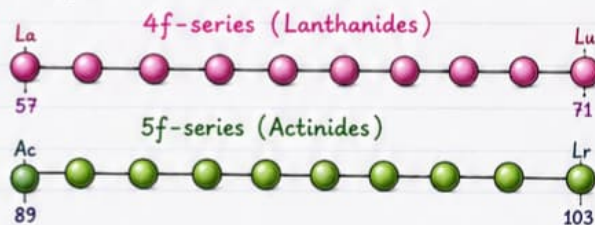
Note

Species with all electrons paired are diamagnetic (e.g., Zn^{2+} , Cu^+).



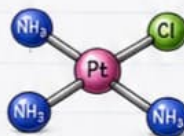
CHARACTERISTIC PROPERTIES OF f-BLOCK ELEMENTS

- All f-block elements are metallic, silvery-white.
- They are soft, malleable and good conductors.
- Most are radioactive.
- They show variable oxidation states.
- They form coloured and complex compounds.



APPLICATIONS

- Fe, Cr, Ni $\rightarrow$ Used in alloys and stainless steel.
- $\text{V}_2\text{O}_5 \rightarrow$ Catalyst in Contact process.
- $\text{MnO}_2 \rightarrow$ Dry cells and decomposing agent.
- $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2] \rightarrow$ Used in anti-cancer drugs.
- Lanthanides $\rightarrow$ Used in colour TV, lasers, magnets.
- Actinides $\rightarrow$ Used in nuclear reactors and nuclear energy.



Square planar complex
 $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$
(Anti-cancer drug - Cisplatin)



IMPORTANT EXAMPLES

- $\text{K}_2\text{Cr}_2\text{O}_7 \rightarrow$ Oxidising agent in acidic medium.
- $[\text{Co}(\text{NH}_3)_6]^{3+} \rightarrow$ Octahedral complex.
- $\text{KMnO}_4 \rightarrow$ Strong oxidising agent.
- $\text{CeO}_2 \rightarrow$ Used in catalyst and glass polishing.
- $\text{UO}_2 \rightarrow$ Nuclear fuel.



d- and f-block elements may be few in number, but their properties and uses are countless!

MASTER THE BASICS, ACE NEET! ♥

END OF
CHAPTER 4